

CMSE 801 - Day 5

More Functions

Sep 19, 2022

General Course Announcements

- Pre-class assignments for Monday are now due Saturday at 5:30 p.m. instead of Friday
- Pre-class assignments for Wednesday are now due Tuesday at 5:30 p.m.
- Homework 1 has been released and is due Wednesday Sep 28 at 11:59 p.m.

Questions/comments from Day 5 PCA

- I just named my function as "print" (just for fun), and it caused all my codes become error. How can I fix it and are there other names we can not use as function name?
- Is there a way to pass undeclared variables to a function. For example, if I pass a list of additional variables, can my function deal with these?
- Just going over the last question, also I was wondering if you could put loops in functions or if we should avoid that
- more exercises on functions. still struggling to grasp how to incorporate loops into functions.
- Can you explain the difference between round and square brackets? My last assignment, each time I used square brackets I always got bad results, but when I change them with round brackets somehow I can get the right answer.
- I think we need more time for in-class assignment during the class.
- I think the third video was confusing, but I guess we should be trying what works and what doesn't too.
- just covering functions more so.
- Last class period you mentioned tears while coding, but for the first time I had tears of joy! Something finally clicked for me quickly!! / Thank you Dr. Law and Jim for being super helpful during office hours! :) / Thank you!

```
In [1]: def my_print():  
        print('allo')  
        a = 1  
        return a  
        my_print()
```

allo

```
Out[1]: 1
```

```
In [2]: def test():  
        return 2*a
```

```
In [3]: b = 1
        test()
```

```
-----
NameError                                Traceback (most recent call last)
<ipython-input-3-812930bc9372> in <module>
      1 b = 1
----> 2 test()

<ipython-input-2-05563f4beb2e> in test()
      1 def test():
----> 2     return 2*a

NameError: name 'a' is not defined
```

```
In [4]: a = 1
        test()
```

Out[4]: 2

```
In [5]: def exam_data(grades):
        nbStudents = len(grades)
        if nbStudents == 0:
            print("There is no data")
        else:
            mean = sum(grades)/nbStudents
            print('mean = ', mean)
            return mean, nbStudents

mean, nbStudents = exam_data([1, 2, 3, 4, 5, 6, 7])
# exam_data([1, 2, 3, 4, 5, 6, 7])
print(mean, nbStudents)
exam_data([])
```

```
mean = 4.0
4.0 7
There is no data
```

```
In [6]: def print_grades(grades):
        for i in grades:
            print(i)

print_grades([1, 2, 3, 4, 5, 6, 7])
```

```
1
2
3
4
5
6
7
```

```
In [7]: def allo(a,b,c):
        print('a =', a)
        print('b =', b)
        print('c =', c)
        allo(1,2,3)
```

```
a = 1
b = 2
c = 3
```

```
In [8]: allo(c=3,b=2.5,a=1)
```

```
a = 1
b = 2.5
c = 3
```

```
In [9]: def name(c=1,a,b):
        print('a =', a)
        print('b =', b)
        print('c =', c)
```

File "<ipython-input-9-cc49d71215c7>", line 1

```
def name(c=1,a,b):
      ^
```

SyntaxError: non-default argument follows default argument

```
In [10]: def name(a,b,c=1,d=2):
        print('a =', a)
        print('b =', b)
        print('c =', c)
        print('d =', d)
        name(3.1, 5.2)
        name(b=3,a=-1)
```

```
a = 3.1
b = 5.2
c = 1
d = 2
a = -1
b = 3
c = 1
d = 2
```

```
In [11]: name(1,2,5,8)
        name(1,2,d=10)
```

```
a = 1
b = 2
c = 5
d = 8
a = 1
b = 2
c = 1
d = 10
```

```
In [12]: def x_squared(vals=[1,2,3]):
        x2 = []
        for x in vals:
            x2.append(x**2)
        return x2

        xsquared = x_squared([2,4,6,8,10])
        print(xsquared)
```

```
[4, 16, 36, 64, 100]
```